

Thur-05-22-2025

We continue our discussions on potential flow

We can show that $\begin{cases} \nabla^2 \psi = 0 \\ \nabla^2 \phi = 0 \end{cases}$ so these are Laplace Eqⁿ for ψ and ϕ

stream function

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Laplace eq.

Substituting eq. (2.35) into eq. (2.44) we get,

$$\nabla^2 \psi = 0 = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \quad (2.45)$$

Hence, when the flow is steady, incompressible, and irrotational (inviscid), both the potential function ϕ and the streamfunction ψ satisfies Laplace's equation (Eq. (2.43) and (2.45) respectively.)

Remember, irrotationality implies that the flow is inviscid

we seek solutions of $\begin{cases} \nabla^2 \phi = 0 \\ \nabla^2 \psi = 0 \end{cases}$ and get $\begin{cases} \phi \\ \psi \end{cases}$ then get velocity by: $\begin{cases} u = \frac{\partial \psi}{\partial y} = \frac{\partial \phi}{\partial x} \\ v = -\frac{\partial \psi}{\partial x} = \frac{\partial \phi}{\partial y} \end{cases}$

Laplace eq.

To summarize if the flow is steady, incompressible & irrotational (inviscid) the stream function ψ (2-D) and the potential function ϕ both satisfy Laplace's equation

$$\nabla^2 \psi = \nabla^2 \phi = 0$$

Such a flow field is referred to as a potential flow field i.e., steady, incompressible & irrotational (inviscid). This is also called ideal flow as such a flow does not exist in nature.

It is a purely mathematical construct

Why is this a big deal?

To recap we have shown that if the flow is steady, incompressible & irrotational (inviscid) the continuity eq. reduces to the Laplace Eq.

(Eq. 2.43 & 2.45) $\rightarrow \nabla^2 \phi = \nabla^2 \psi = 0$. $\xrightarrow{\text{linear}}$

- ▶ 1 equation, 1 unknown $\rightarrow$ Euler's eq. 4 eqns. 4 unknowns
- ▶ Linear equation $\rightarrow$ can use principle of superposition $\rightarrow$ if $\phi_1, \phi_2, \dots, \phi_n$ are all solutions to a linear equation then the linear combination of any of the solutions are also a solution. e.g., $\phi_i + 1/3\phi_j + 5\phi_k$ is also a solution. We shall exploit this. We shall find some basic solutions to the Laplace equation and then find linear combinations of them to yield other solutions.

Mem. Eq: Non-linear

$\left\{ \begin{array}{l} 1 \text{ continuity} \\ 3 \text{ Mom} \end{array} \right.$

Bernoulli's Eq. (TSec. 3.2)

Lets start with the steady, two-dimensional, inviscid momentum equations without body forces (eq. (2.5)) and the continuity equation (eq. (1.12))

$$\text{Continuity: } \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = \vec{\nabla} \cdot (\rho \vec{V}) = 0 \quad (2.46)$$

Conservative form
of Mom. Eq.

$$\text{x-momentum: } \vec{\nabla} \cdot (\rho \vec{V} u) = -\frac{\partial p}{\partial x} \quad (2.47)$$

$$\text{y-momentum: } \vec{\nabla} \cdot (\rho \vec{V} v) = -\frac{\partial p}{\partial y} \quad (2.48)$$

$$\text{x-Mom} \longrightarrow \vec{\nabla} \cdot (s \vec{V} u) = \cancel{s u \vec{\nabla} \cdot \vec{V}} + s \vec{V} \cdot \vec{\nabla} u$$

$$\implies \vec{\nabla} \cdot (s \vec{V} u) = s \vec{V} \cdot \vec{\nabla} u$$

$$\implies \text{x-Mom} \longrightarrow s \vec{V} \cdot \vec{\nabla} u = -\frac{\partial p}{\partial x} \quad (\text{Non-conservative form of Mom. Eq.})$$

Governing eqn.

Also observe the following vector identity,

$$\nabla \cdot (s \vec{F}) = s (\nabla \cdot \vec{F}) + \vec{F} \cdot (\nabla s) \quad (2.49)$$

where, s is any scalar and $\vec{F}$ any vector. Apply this identity to the momentum equations and then use the fact that from the steady continuity $\rightarrow \nabla \cdot (\rho \vec{V}) = 0$. Therefore, the momentum equations become:

Non-conservative form of Mom. $\left\{ \begin{array}{l} \text{x-momentum: } \rho \vec{V} \cdot (\nabla u) = -\frac{\partial p}{\partial x} \quad (2.50) \\ \text{y-momentum: } \rho \vec{V} \cdot (\nabla v) = -\frac{\partial p}{\partial y} \quad (2.51) \end{array} \right.$

Therefore the density can be taken out of the gradient operator without having to assume incompressibility. It's been shown for the steady equations, also can be shown for the unsteady equations.

x-Mom: $\rho \vec{V} \cdot \vec{\nabla} u = -\frac{\partial p}{\partial x}$ (vector form)

$$\rho (u \hat{i} + v \hat{j}) \cdot \left(\frac{\partial u}{\partial x} \hat{i} + \frac{\partial u}{\partial y} \hat{j} \right) = \rho \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] = -\frac{\partial p}{\partial x}$$

Expanded form

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad \times dx$$

Bernoulli's eq.

Eq. (2.50) expanded out is,

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

Multiply eq. (2.52) by dx :

$$u \frac{\partial u}{\partial x} dx + v \frac{\partial u}{\partial y} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

$$u \frac{\partial u}{\partial x} dx + v \frac{\partial u}{\partial y} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx \quad (2.52)$$

Irrrot flow: $\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$

(2.53)

Consider irrotational flow, i.e., $\vec{\nabla} \times \vec{V} = 0$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = 0 \quad 2-D \rightarrow \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$

(2.54)

$$\Rightarrow u \frac{\partial u}{\partial x} dx + v \frac{\partial v}{\partial x} dx = -\frac{1}{\rho} \frac{\partial p}{\partial x} dx$$

Bernoulli's eq.

Substituting eq. (2.54) into eq. (2.53) we get,

$$u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (2.55)$$

Similarly, from the y-momentum we get,

$$u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (2.56)$$

Adding eq. (2.55) and 2.56,

$$\underbrace{u \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right)}_{du} + \underbrace{v \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right)}_{dv} = -\frac{1}{\rho} \left(\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy \right) \quad (2.57)$$

Bernoulli's eq.

Remember, by chain rule

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad (2.58)$$

Therefore, eq. (2.57) becomes,

$$d\left(\frac{U^2}{2}\right) = \boxed{u \, du + v \, dv = -\frac{1}{\rho} dp} \quad (2.59)$$

$$U = |\vec{V}| = \sqrt{u^2 + v^2} \quad (2.60)$$

$$U^2 = |\vec{V}|^2 = u^2 + v^2 \quad (2.61)$$

$$d(U^2) = d(u^2 + v^2) = 2u \, du + 2v \, dv \quad (2.62)$$

$$d\left[\frac{U^2}{2}\right] = u \, du + v \, dv \quad (2.63)$$

Compressible Bernoulli's eq.

Comparing eq. (2.63) in eq. (2.59) we get,

$$\boxed{d\left[\frac{U^2}{2}\right] = -\frac{dp}{\rho}} \rightarrow \text{Compressible Bernoulli's eq.} \quad (2.64)$$

Assumptions:

- steady $\frac{\partial}{\partial t} = 0$
- inviscid $\mu = 0$
- irrotational $\nabla \times \vec{V} = 0$

Also in this form, it is assumed that there are no body forces and can be included as in your text.

To solve we integrate Eq. 2.64

Incompressible Bernoulli's eq.

If the flow is incompressible, $\rho \approx \text{constant}$ we can integrate eq. 2.64

$$\int \frac{dp}{\rho} = - \int d\left[\frac{U^2}{2}\right] \quad (2.65)$$

$$\frac{p}{\rho} + \frac{U^2}{2} = \text{const} \quad (2.66)$$

Or it is rewritten more commonly in the form,

Bernoulli's Eqⁿ $\boxed{p + \frac{1}{2}\rho U^2 = \text{const}} \quad (2.67)$

Also in this form it is assumed that there are no body forces and can be included as in your text.

Incompressible Bernoulli's eq.

$$p + \frac{1}{2}\rho U^2 = \text{const}$$

Assumptions:

- ▶ steady ✓
- ▶ inviscid ✓
- ▶ irrotational ✓
- ▶ incompressible ✓

Remember, these assumptions collectively are the potential flow assumptions.

$$C_p = \frac{P - P_\infty}{\frac{1}{2} \rho U_\infty^2}$$

Pressure
coeff.

C_p (TSec. 3.4)

$$C_p = \frac{p - p_\infty}{0.5 \rho_\infty U_\infty^2} \quad (2.68)$$

When the flow is potential i.e., steady, incompressible, irrotational (inviscid), and when we ignore body forces, we can apply Bernoulli's equation between a point in the flow and a point in the free-stream.

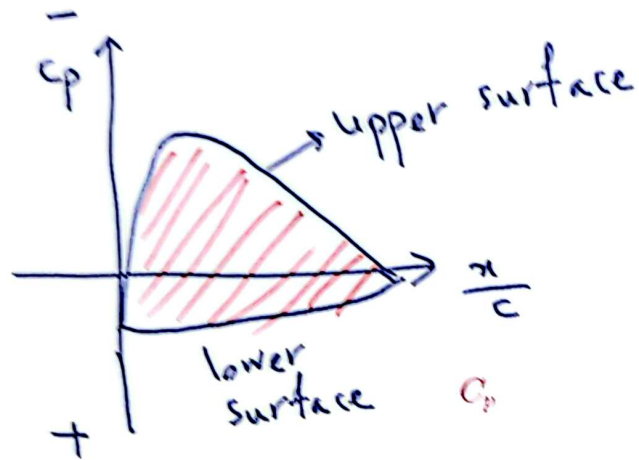
$$p + \frac{1}{2} \rho_\infty U^2 = p_\infty + \frac{1}{2} \rho_\infty U_\infty^2 \quad (2.69)$$

$$p - p_\infty = \frac{1}{2} \rho_\infty U_\infty^2 \left[1 - \left(\frac{U}{U_\infty} \right)^2 \right] \quad (2.70)$$

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$$\left\{ \frac{P - P_\infty}{\frac{1}{2} \rho U_\infty^2} = \left[1 - \left(\frac{U}{U_\infty} \right)^2 \right] = C_p \right.$$

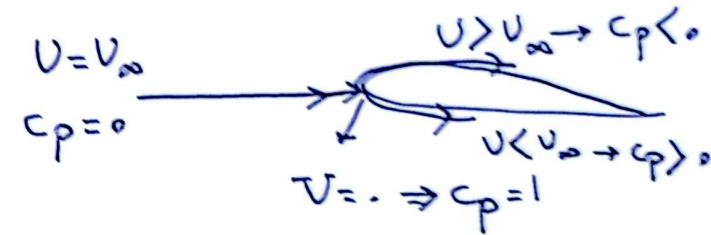
"If velocity is given, you can compute C_p from this relation"



C_p over airfoil

$$C_L = \int_0^c (C_{p_l} - C_{p_u}) dx$$

↓
lift coef.



Therefore, when the flow is steady, incompressible & irrotational (inviscid) (together these are the potential flow assumptions) and when we ignore body forces

$$C_p = 1 - \left(\frac{U}{U_\infty} \right)^2 \quad (2.71)$$

$$\text{At a stagnation point: } \rightarrow U = 0 \rightarrow C_p = 1 \quad (2.72)$$

$$\text{At a point where } \rightarrow U = U_\infty \rightarrow p = p_\infty \rightarrow C_p = 0 \quad (2.73)$$

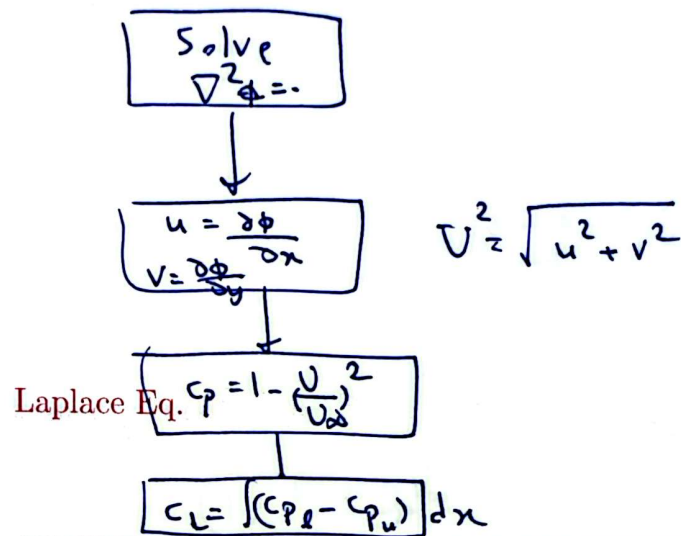
$$\text{If } p > p_\infty \rightarrow U < U_\infty \rightarrow C_p > 0 \quad (2.74)$$

$$\text{If } p < p_\infty \rightarrow U > U_\infty \rightarrow C_p < 0 \quad (2.75)$$

Solving Potential Flows (TSec. 3.11 & 3.8.2)

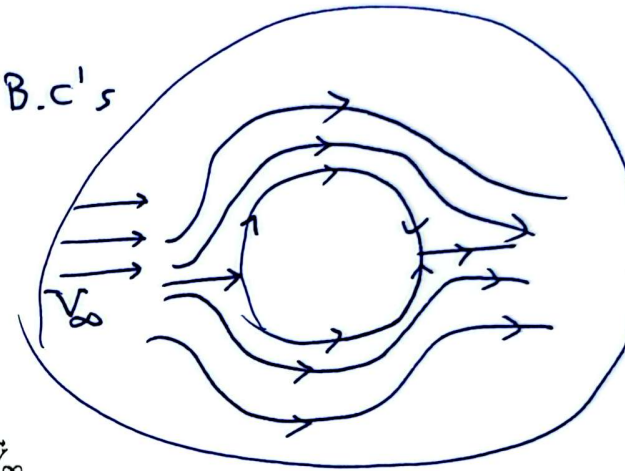
To recap we have shown that if the flow is potential i.e., steady, incompressible & irrotational (inviscid) the continuity eq. reduces to the Laplace Eq. (Eq. (2.43) & (2.45)) $\rightarrow \nabla^2 \phi = \nabla^2 \psi = 0$.

- ▶ 1 equation, 1 unknown $\rightarrow$ Euler's eq. 4 eqns. 4 unknowns
- ▶ Linear equation $\rightarrow$ can use principle of superposition $\rightarrow$ if $\phi_1, \phi_2, \dots, \phi_n$ are all solutions to a linear equation then the linear combination of any of the solutions is also a solution. e.g., $\phi_i + 1/3\phi_j + 5\phi_k$ is also a solution. We shall exploit this. We shall find some basic solutions to the Laplace equation and then find linear combinations of them to yield other solutions.

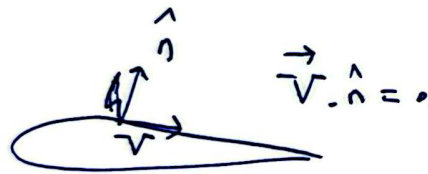


- ▶ Remember, $\nabla^2 \phi = \nabla^2 \psi = 0$, is an expression of the conservation of mass for potential i.e., steady, incompressible & irrotational (inviscid) flow. Also 2-d in the case of ψ .
- ▶ So what happens to the momentum equations? → Bernoulli's equation $p/\rho + U^2/2 = \text{const}$, everywhere. We use $\nabla^2 \phi = \nabla^2 \psi = 0$ to solve for u and v and then use Bernoulli's eq. to calculate the pressure
- ▶ If all potential flows were governed by Laplace and Bernoulli's eqns. how do we distinguish between flows? for example, flow over an airfoil and that over a cylinder? → boundary conditions

$\nabla^2 \phi = 0 \rightarrow$ needs 2 B.C's



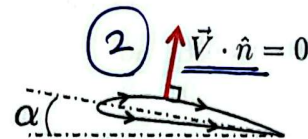
Boundary conditions



Body is a streamline, i.e., velocity is tangent to the body.

① $\vec{V} @ \infty = \vec{V}_\infty$

$\vec{V} @ \infty = \vec{V}_\infty$



$\vec{V} @ \infty = \vec{V}_\infty$

$\vec{V} @ \infty = \vec{V}_\infty$

Figure 2.13: Boundary conditions for potential flow over an airfoil. Based on image from FA-JA

Boundary conditions

Far away from the body (∞),

$$\vec{V} = \vec{U}_{\infty} \quad (2.76)$$

For the airfoil shown this is expressed as,

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = U_{\infty} \cos \alpha \quad (2.77)$$

$$v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} = U_{\infty} \sin \alpha \quad (2.78)$$



$$u \text{ at free stream: } U_{\infty} \cos \alpha = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$$

$$v \text{ at " " } U_{\infty} \sin \alpha = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

B.c for potential flow over the body : $\vec{V} \cdot \hat{n} = 0$

" " Viscous ζ ζ ζ ζ ζ $u = u_{wall}$
 $v = v_{wall}$

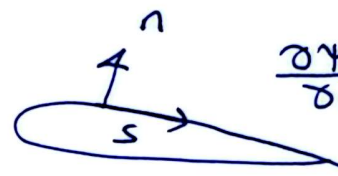
Boundary conditions

No-slip

At the wall for potential flows we need to be careful. Consider the 2-D incompressible Navier-Stokes equations – continuity and momentum only. We have 3 equations (continuity, x -momentum and y -momentum) and 3 unknowns (p, u, v). So while solving these set of equations, on the wall of the airfoil shown above (and analogously any body in a real flow) we can enforce $p = p_{wall}$, $u = u_{wall} = 0$ & $v = v_{wall} = 0$. Note, $u = u_{wall} = 0$ & $v = v_{wall} = 0$ together is called the no-slip boundary condition. However, in potential flows we have only one equation to solve for both u and $w \rightarrow \nabla^2 \psi = 0$ or $\nabla^2 \phi = 0$. Hence, we can only impose one boundary condition at the wall.

We say that the flow cannot penetrate the surface of the airfoil or there is no flow through the airfoil surface

$$\begin{cases} \vec{V} \cdot \vec{n} = 0 \\ \vec{V} = \vec{\nabla} \phi \end{cases} \Rightarrow \vec{\nabla} \phi \cdot \vec{n} = 0 \quad \text{or} \quad \frac{\partial \phi}{\partial n} = 0 = \frac{\partial \psi}{\partial s}$$


 $\frac{\partial \psi}{\partial s} = 0 \rightarrow \psi = \text{cte}$
 over body
 $\Rightarrow$ body is streamline

Boundary conditions

No flow through the surface i.e.,

$$\vec{V} \cdot \vec{n} = 0 \quad (2.79)$$

Note, for potential flow

$$\vec{V} = \vec{\nabla} \phi \rightarrow \vec{\nabla} \phi \cdot \vec{n} = 0 \quad (2.80)$$

In terms of the stream function

$$\frac{\partial \psi}{\partial s} = 0 \quad (2.81)$$

Therefore, on the surface of the body

$$\delta \psi = 0 \rightarrow \psi = \text{const} \quad (2.82)$$

Therefore, the surface of the body (airfoil) is a streamline.

$$\nabla^2 \phi = \nabla^2 \psi = 0$$

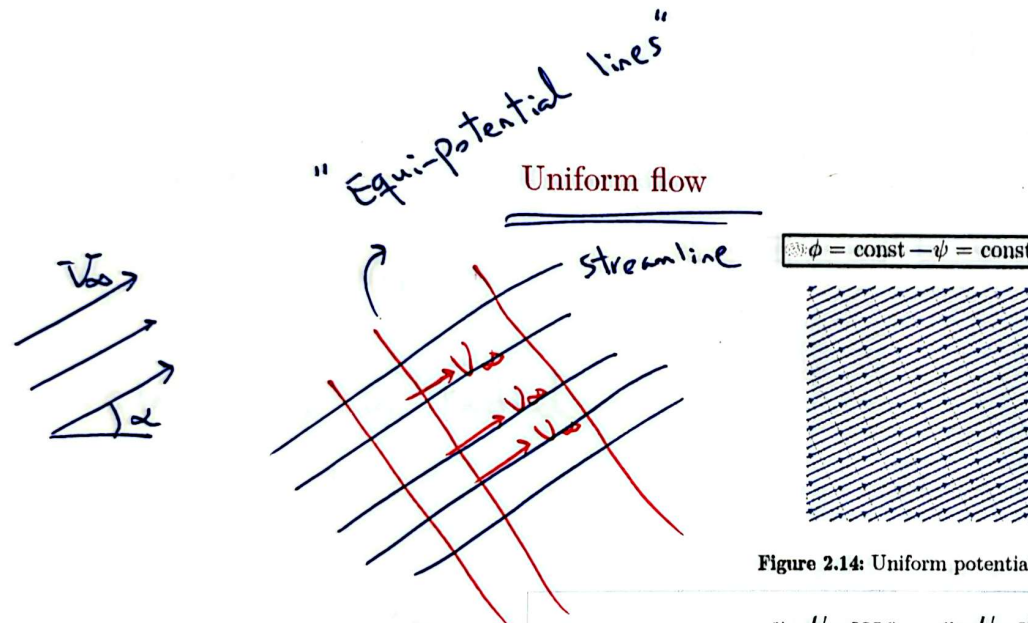


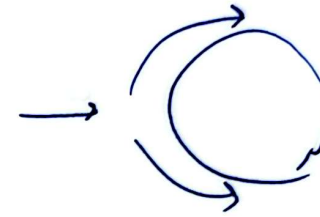
Figure 2.14: Uniform potential flow

In Cart.

In polar coord.

$$\begin{aligned} u &= U_{\infty} \cos \alpha & v &= U_{\infty} \sin \alpha & (2.83) \\ \phi &= U_{\infty} (x \cos \alpha + y \sin \alpha) & \psi &= U_{\infty} (y \cos \alpha - x \sin \alpha) & (2.84) \\ \phi &= U_{\infty} r (\cos \theta \cos \alpha + \sin \theta \sin \alpha) & \psi &= U_{\infty} r (\sin \theta \cos \alpha - \cos \theta \sin \alpha) & (2.85) \end{aligned}$$

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$$\begin{aligned} u &= U_{\infty} \sin \alpha = \frac{\partial \phi}{\partial x} \\ v &= U_{\infty} \cos \alpha = \frac{\partial \phi}{\partial y} \end{aligned}$$

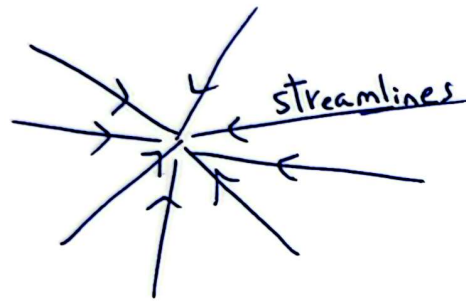
$$\phi = U_{\infty} x \sin \alpha + k(y)$$

$$\frac{\partial \phi}{\partial y} = \frac{dk}{dy} = U_{\infty} \cos \alpha$$

$$\rightarrow k = U_{\infty} \cos \alpha y + C$$

$$\Rightarrow \phi = U_{\infty} (x \sin \alpha + y \cos \alpha) + C$$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 + 0 = 0$$



Sink

(Source)

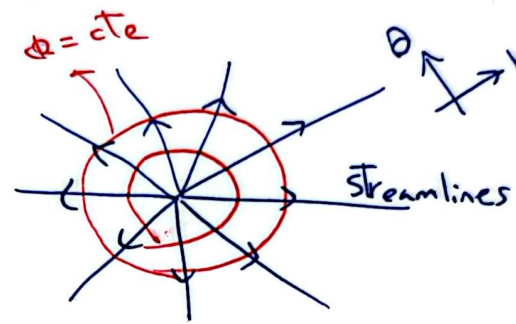
$$V_r = \frac{k}{2\pi r}$$

$$V_\theta = 0$$

velocity is in radial direction!

velocity decays as distance from origin increases

$$V_r \propto \frac{1}{r}$$



Source

Source/sink flow

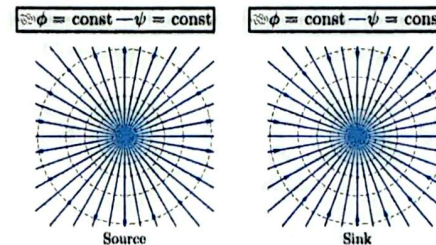


Figure 2.15: Source/sink flow

$$v_r = \frac{K}{2\pi r} \quad v_\theta = 0 \quad (2.86)$$

$$\phi = \frac{K}{2\pi} \ln r \quad \psi = \frac{K}{2\pi} \theta \quad (2.87)$$

$$K > 0 \text{ source} \quad K < 0 \text{ sink} \quad (2.88)$$

Note: θ is in radians, and the source and sink are located at the origin.

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$$\frac{k}{2\pi r} = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$$

$$\Rightarrow \psi = \frac{k}{2\pi} \theta$$

(Sink)

$$V_r = \frac{-k}{2\pi r}$$

$$V_r = \frac{\partial \phi}{\partial r} = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$$

$$\frac{k}{2\pi r} = \frac{\partial \phi}{\partial r} = \frac{d\phi}{dr}$$

$$\phi = \frac{k}{2\pi} \ln r$$

Derived flow - doublet flow

We are going to use the principle of superposition to derive our third flow - a doublet. Consider a source and a sink as shown below.

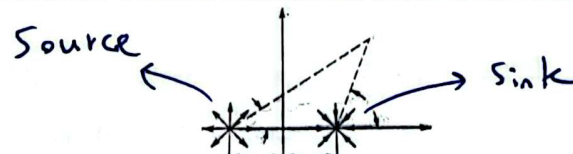


Figure 2.16: doublet potential flow, adapted from FA-KC

The stream function for the entire flow field is now

$$\psi = \psi_{source} + \psi_{sink} \quad (2.89)$$

$$= -\frac{K}{2\pi} (\theta_1 - \theta_2) = -\frac{K}{2\pi} \left(\tan^{-1} \frac{y}{x-x_0} - \tan^{-1} \frac{y}{x+x_0} \right) \quad (2.90)$$

Derived flow - doublet flow

$$\tan^{-1} \alpha - \tan^{-1} \beta = \tan^{-1} \left(\frac{\alpha - \beta}{1 + \alpha\beta} \right) \quad (2.91)$$

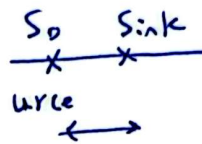
$$\text{After some algebra, } \psi(x, y) = -\frac{K}{2\pi} \tan^{-1} \frac{2x_0 y}{x^2 + y^2 - x_0^2} \quad (2.92)$$

$$\text{let, } B = 2x_0 K \rightarrow \text{doublet strength when } x_0 \rightarrow 0 \quad (2.93)$$

$$\text{therefore, } \psi_{\text{doublet}} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \right] \quad (2.94)$$

$$x_0 \rightarrow 0, \frac{2x_0 y}{x^2 + y^2 - x_0^2} \rightarrow \text{something small} \quad (2.95)$$

$$\text{therefore as } x_0 \rightarrow 0, \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \rightarrow \frac{2x_0 y}{x^2 + y^2 - x_0^2} \quad (2.96)$$

Doublet $\rightarrow$ 

$$\psi_{\text{doublet}} = \frac{-B}{2\pi} \frac{\sin\theta}{r}$$

Derived flow - doublet flow

$$\text{therefore, } \psi_{\text{doublet}} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \frac{2x_0 y}{x^2 + y^2 - x_0^2} \right] \quad (2.97)$$

$$\rightarrow -\frac{B}{2\pi} \frac{y}{x^2 + y^2} = -\frac{B \sin\theta}{2\pi r} \quad (2.98)$$

Doublet $\rightarrow$

$$v_r = -\frac{B \cos\theta}{2\pi r^2} \quad v_\theta = -\frac{B \sin\theta}{2\pi r^2} \quad (2.99)$$

$$\phi = \frac{B \cos\theta}{2\pi r} \quad \psi = -\frac{B \sin\theta}{2\pi r} \quad (2.100)$$

Note, Even though we derived this flow, now that we have the final result i.e., ψ and ϕ , the doublet also becomes an elementary flow. We can just use it, by adding it to other flows. This is the power of linear models!

Derived flow - doublet flow

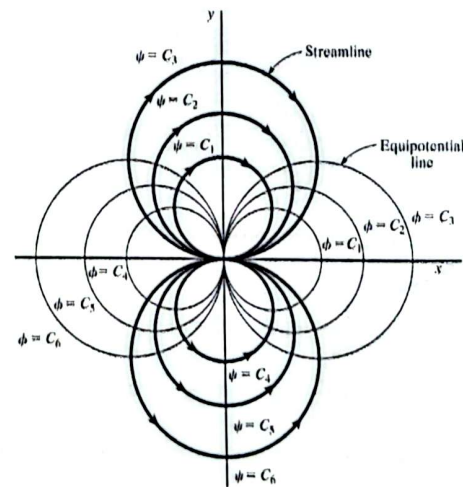


Figure 2.17: Fig. 3.11 from AE-BC

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Source : $V_r \propto \frac{1}{r}$, $V_\theta = 0$

Vortex : $V_r = 0$, $V_\theta \propto \frac{1}{r}$ $\left\{ \begin{array}{l} \text{streamline: "circle"} \\ \text{Eq. pot: "radial"} \end{array} \right.$

4) Vortex flow

$$\phi = \text{const} \rightarrow \psi = \text{const}$$

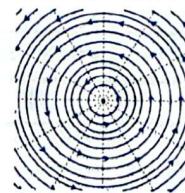


Figure 2.18: Vortex potential flow

$$v_r = 0 \quad v_\theta = -\frac{\Gamma}{2\pi r} \quad (2.101)$$

$$\phi = -\frac{\Gamma}{2\pi} \theta \quad \psi = \frac{\Gamma}{2\pi} \ln r \quad (2.102)$$

Note: θ is in radians, as in the source and the sink Γ is centered at the origin.

Summary

TABLE 3.3 Stream Functions and Potential Functions for Elementary Flows		
Flow	ψ	ϕ
Uniform flow in x direction	$U_\infty r \sin \theta, U_\infty y$	$U_\infty r \cos \theta, U_\infty x$
Uniform flow at angle α	$U_\infty (y \cos \alpha - x \sin \alpha)$	$U_\infty (x \cos \alpha + y \sin \alpha)$
Source	$\frac{K\theta}{2\pi}$	$\frac{K}{2\pi} \ln r$
Doublet	$-\frac{B}{r} \sin \theta$	$\frac{B}{r} \cos \theta$
Vortex (with clockwise circulation)	$\frac{\Gamma}{2\pi} \ln r$	$-\frac{\Gamma\theta}{2\pi}$
90° corner flow	Axy	$\frac{1}{2}A(x^2 - y^2)$
Solid-body rotation	$\frac{1}{2}\omega r^2$	Does not exist since flow is rotational

Figure 2.19: Table 3.3 from AE-BC

$$R = \sqrt{\frac{B}{2\pi U_\infty}}$$

Non-spinning cylinder (TSec. 3.13)

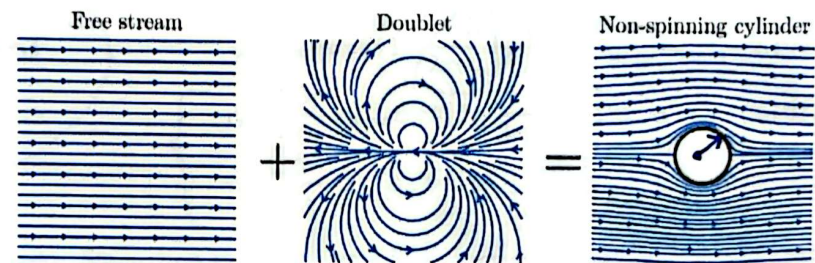


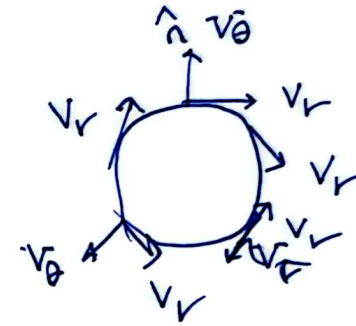
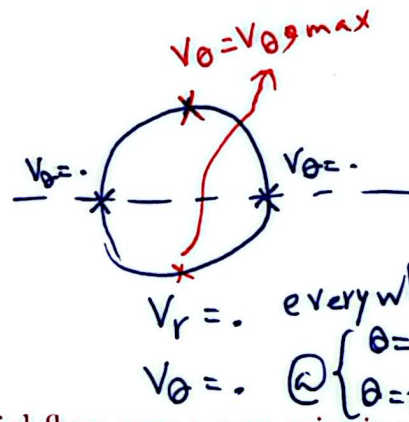
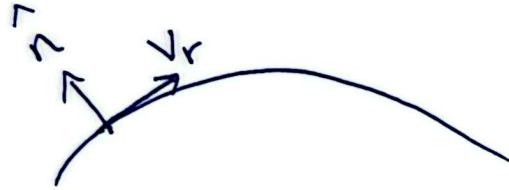
Figure 2.20: Potential flow over a non-spinning cylinder. Based on image from FA-JA

Non lifting
" Dragging

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} \quad (2.103)$$

$$= U_\infty r \sin \theta + \frac{-B \sin \theta}{2\pi r} \quad (2.104)$$

$$\psi = U_\infty r \sin \theta \left(1 - \frac{B}{2\pi U_\infty r^2} \right) \quad (2.105)$$



Potential flow over a non-spinning cylinder

stagnation means

$$V_r = V_\theta = 0$$

$$V_r = 0 \rightarrow r = R$$

$$V_\theta = 0 \text{ if } \theta = 0$$

$$\theta = \pi$$

$$(\sin \theta) = 0$$

C_p over
the cylinder



Stagnation pt. $|\vec{V}| = 0 \rightarrow r = R = \sqrt{\frac{B}{2\pi U_\infty}}$

$$C_p = 1 - \left(\frac{U}{U_\infty}\right)^2 = 1 - \left(\frac{\sqrt{v_r^2 + v_\theta^2}}{U_\infty}\right)^2$$

$$C_p = 1 - 4\sin^2 \theta$$

$$\psi = U_\infty r \sin \theta \left(1 - \frac{B}{2\pi U_\infty r^2}\right) \quad (2.106)$$

$$\left\{ \begin{aligned} v_r &= \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_\infty \cos \theta \left[1 - \frac{B}{2\pi U_\infty r^2}\right] = 0 \\ v_\theta &= -\frac{\partial \psi}{\partial r} = -U_\infty \sin \theta \left[1 + \frac{B}{2\pi U_\infty r^2}\right] \end{aligned} \right. \quad (2.107)$$

$$v_\theta = -\frac{\partial \psi}{\partial r} = -U_\infty \sin \theta \left[1 + \frac{B}{2\pi U_\infty r^2}\right] \quad (2.108)$$

$$r = R \quad (2.109)$$

$$C_p = 1 - 4\sin^2 \theta \quad (2.110)$$

$$C_p = 1 - 4\sin^2 \theta \quad (2.111)$$

$$\vec{V} \cdot \hat{n} = 0$$

$$\vec{V} = V_r \hat{e}_r + V_\theta \hat{e}_\theta$$

$$\hat{n} = \hat{e}_\theta$$

$$V_r = 0 \Rightarrow S_o$$

$$\underline{V_r \cdot \hat{n} = V_r \cdot \hat{e}_\theta = 0}$$

$$V_r = 0 \Rightarrow 1 - \frac{B}{2\pi U_\infty r^2} = 0$$

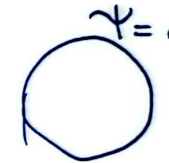
$$r^2 = \frac{B}{2\pi U_\infty} \text{ or } r = \underbrace{\sqrt{\frac{B}{2\pi U_\infty}}}_R$$

$$R = \sqrt{\frac{B}{2\pi U_\infty}}$$

Potential flow over a non-spinning cylinder

$$\psi = U_\infty r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] \quad (2.112)$$

$$\psi(r=R) = 0 \rightarrow \text{surface streamline} \quad (2.113)$$



Note, that it just happens that $\psi = 0$ is the surface streamline. There is no other significance to this.

- Stagnation point $\rightarrow \theta = 0 \text{ \& } \pi, r = R$. (Show as an ungraded homework)
- Max velocity $\rightarrow r = R, \theta = \pi/2 \text{ \& } 3\pi/2, v_r = 0, v_\theta = 2U_\infty$ (Show as an ungraded homework)
- At $r = R, v_r = 0$ and $v_\theta = -2U_\infty \sin \theta$ (Show as an ungraded homework)

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$$v_\theta = -2U_\infty \sin \theta$$

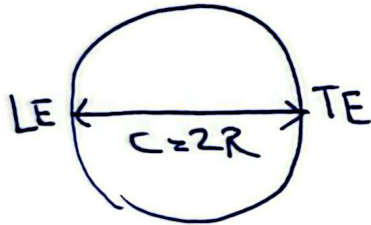
over
the
cyl ($r=R=\sqrt{\frac{B}{2\pi U_\infty}}$)

$$v_r = 0$$

$$C_d = \frac{1}{c} \int_{LE}^{TE} (C_{pu} - C_{pl}) dy = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta$$

chord : $c = 2R$

$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta$$



Potential flow over a non-spinning cylinder

- $C_l = \frac{1}{c} \int_{le}^{te} (C_{pl} - C_{pu}) dy = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is top/bottom symmetry hence there is no net force in the y direction and thus no lift.
- $C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dx = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.

$$C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dy \quad (2.114)$$

$$y = R \sin \theta \rightarrow dy = R \cos \theta d\theta, \quad c = 2R \quad (2.115)$$

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta \quad \text{Similarly } C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta \quad (2.116)$$